TRACEGATE

Invention development brief: configuration-linked autonomy verification and change-impact gating

Prepared 19 October 2026
Source basis: four user-provided patent development packages

Decision-support package for selecting the next system or patent concept to develop.

STATUS | Portfolio synthesis and concept hypotheses only
No novelty, freedom-to-operate, engineering-performance, or filing-readiness conclusion is implied.

1. Concept at a glance

Working name only TRACEGATE is a provisional project label, not a trademark clearance or patent title recommendation.

TRACEGATE is a proposed engineering assurance system that links a specific autonomous-system configuration to its safety requirements, assumptions, test scenarios, and evidence records. When a hardware, sensor, payload, calibration, software, model, or approved operating limit changes, it determines which proof obligations are affected and which validation artifacts must be refreshed before the changed configuration can receive a bounded readiness status.

Scope boundary: TRACEGATE is primarily a pre-deployment, maintenance-change, and software-release validation system. It does not plan or command vehicle motion, allocate tactical missions, select targets, or replace independent real-time safety controllers.

2. Draft problem statement

Autonomous robotic programs commonly maintain requirements, configuration records, hazards, simulation tests, field-test reports, software/model versions, and maintenance data in separate artifacts. When an item changes, engineers may rerun too few tests (leaving stale evidence) or rerun every test (wasting time) without an auditable link between the change, impacted safety assumptions, test selection, and release decision. This problem is visible across the source portfolio: each package has its own safety/validation appendix, while the cross-package link from configuration change to specific evidence refresh is not defined as a shared mechanism.

3. Fit with existing portfolio

Existing package	What TRACEGATE reuses	What it should not duplicate
AEROSYNC	Mission/state metadata, fault-injection scenarios, test evidence, and vehicle configuration identifiers.	The swarm allocator, flight controller, or runtime task reassignment logic.
AQUAVANTIS-G (currently a UGV package)	Payload identity, sensor validity, localization/terrain assumptions, recovery assumptions, and safe-state test cases.	The UGV mission-feasibility planner or drive safety supervisor.
SAGE-APC	Versioned physical module identity, installation evidence, condition, crew-safety constraints, and inspection states.	The physical armor/module design or any claim that a digital record certifies protective performance.
UCAS revised package	Model identity, calibration/provenance, safety-envelope evidence, health assumptions, reconciliation events, and evidence logs.	Flight trajectory selection, onboard AI planning, or the deterministic flight safety gate.

4. Proposed architecture

Component	Inputs	Function and outputs
A. Configuration snapshot	Hardware/part IDs and revisions;	Creates an immutable, versioned baseline

Component	Inputs	Function and outputs
	sensors/calibration; payload or module set; software/model versions; maintenance state; operating domain.	and marks required fields, trusted sources, timestamps, and unresolved identity/configuration conflicts.
B. Requirement and assumption graph	Safety requirements, hazards, operating limits, environmental assumptions, test cases, acceptance criteria, and evidence IDs.	Stores typed dependencies: what claims depend on which components, assumptions, tests, and evidence conditions.
C. Change-impact engine	Typed change event and before/after configuration snapshots.	Traverses dependencies, finds affected requirements/assumptions/evidence, and marks old evidence valid, conditional, stale, contradicted, or missing.
D. Test-selection planner	Impact graph, safety criticality, uncertainty, prior failures, test cost/availability.	Selects a justified validation subset and records why each test is selected, retained, or excluded. Safety-critical omissions require an explicit rule or human review.
E. Reproducible test runner/adapters	Approved simulation, bench, replay, or supervised field-test interfaces.	Runs or dispatches tests; captures environment, software/model revision, configuration fingerprint, inputs, outputs, uncertainty, and acceptance results.
F. Evidence ledger	Test outputs, reviewer dispositions, measurement provenance, hashes/signatures where appropriate.	Binds each evidence item to the exact configuration and assumptions under which it was produced; supports audit and repeatability.
G. Readiness decision gate	Completion of proof obligations, evidence freshness/applicability, conflicts, and review rules.	Outputs READY, READY WITH RESTRICTIONS, BLOCKED/UNKNOWN, or REVIEW REQUIRED for a defined use envelope, with reason codes and evidence links.
H. Change/field feedback loop	Maintenance events, new failure cases, anomalies, expiring assumptions, and approved rule updates.	Adds regression scenarios and can revoke or reopen an earlier readiness result when its evidence basis changes.

5. Proposed operating workflow

1. Register and authenticate the baseline configuration; verify that the declared physical and software inventory is complete enough for the intended use.
2. Load the applicable requirement, hazard, operating-domain, and acceptance-rule set. Mark any assumptions that are outside scope or lack evidence.
3. Receive a typed change event (for example: sensor replacement, calibration update, payload change, model upgrade, library update, or operating-envelope change).
4. Compare the old and new configuration snapshots and traverse the dependency graph to identify impacted requirements and evidence.
5. Select required tests under deterministic rules, with uncertainty and safety criticality recorded. Where impact is ambiguous or critical evidence is missing, require manual review rather than assuming pass.
6. Execute, replay, or dispatch each required test and bind the result to exact configuration and environment records.
7. Evaluate readiness rules. Do not issue an unqualified READY status if any mandatory evidence is missing, stale, contradictory, or outside its applicable domain.

8. Publish an audit package containing the proposed change, impacted proof obligations, executed and skipped tests with rationale, result states, approver, and rollback/review conditions.

6. Example use case

Example: a civilian inspection quadrotor receives a new camera and a revised perception model. TRACEGATE does not decide where the aircraft flies. It compares the new camera/calibration/model snapshot with the approved baseline, traverses links to affected perception-quality, time-synchronization, localization-confidence, data-quality, compute-latency, and safe-stop requirements, and identifies which simulation/replay/bench tests are no longer supported by current evidence. It retains unrelated evidence only if its applicability conditions remain satisfied. If a required test is unavailable or fails, the result is BLOCKED/UNKNOWN or READY WITH RESTRICTIONS, with the reason and outstanding proof obligations exposed for human disposition.

A second, physical-configuration example can use SAGE-APC's modular configuration ledger: a module move or service event could trigger tests for identity, installation evidence, relevant crew-safety constraints, sensor validity, and inspection status without suggesting that a software status alone certifies armor performance.

7. Candidate technical differentiation hypotheses

These are hypotheses to test in a claim-focused search; they are not conclusions of novelty or non-obviousness.

- A typed dependency graph that explicitly joins physical configuration items, software/model/calibration versions, safety assumptions, test cases, acceptance rules, and evidence validity conditions.
- Event-driven propagation from a specific configuration change to a deterministic list of invalidated, conditionally valid, and still-valid proof obligations, with an explanatory trace.
- Selection of a targeted validation set whose inclusion/exclusion rationale is tied to dependency reachability, requirement criticality, uncertainty, and known failure history rather than a generic risk score.
- A fail-aware readiness state machine that can downgrade or revoke status when evidence expires, conflicts, or no longer matches the configuration or defined operating domain.
- A reproducible evidence package that captures the configuration fingerprint, applicable assumptions, selected tests, test outputs, skipped-test rationale, reviewer disposition, and conditions for revalidation.

Avoid as the central claim Generic phrases such as 'use AI to make a robot safer,' 'generate a risk score,' 'store configuration data,' 'use a digital twin,' or 'run regression tests' are too broad to distinguish the intended mechanism without more specific technical implementation.

8. Candidate claim map for counsel discussion

Potential claim family	Focus to investigate	Support required
System claim	A processor and data store implementing configuration snapshots, typed dependencies, change-impact propagation, targeted proof-obligation selection, evidence applicability checks, and a bounded readiness state machine.	Detailed graph schema; traversal rules; state machine; specific examples; explanation of technical effect.

Potential claim family	Focus to investigate	Support required
Method claim	Receiving a configuration change; identifying affected requirements and evidence; computing required tests; collecting results; updating readiness and invalidating unsupported approvals.	End-to-end flow, exact decision rules, alternate embodiments, error handling.
Computer-readable medium	Executable instructions performing the specific event-driven impact and evidence-refresh workflow.	Implementation detail sufficient to show a real technical process rather than a result-only abstraction.
Dependent claims	Evidence expiration/applicability, unresolved conflict handling, human review, test-selection rationale, version binding, rollback, and cross-platform adapters.	Each feature should be actually conceived, enabled, and supported by a figure/example.

Do not file claims copied from this roadmap without verifying actual conception and support. Inventorship depends on contribution to claimed subject matter, not on the person who assembles documents.

9. Prototype and validation plan

Test	Method	Measurement / exit evidence
Change-impact coverage	Create a labeled set of configuration changes and known impacted requirements/tests.	Recall of impacted proof obligations; false-negative count for safety-critical items.
Evidence freshness	Inject expired, stale, mismatched-version, missing, contradictory, and out-of-domain records.	No unqualified READY state where mandatory evidence is invalid; reason codes are correct.
Targeted test efficiency	Compare TRACEGATE's required test set with a full-regression baseline and an expert-selected baseline.	Number/cost of tests selected; uncovered known dependencies; rationale reproducibility.
Reproducibility	Re-run the same scenario under fixed configuration and data; vary only declared factors.	Stable selection and status outputs, with provenance sufficient to explain differences.
Change revocation	Change a previously approved component, assumption, or acceptance rule.	Previous readiness is reopened or downgraded when its supporting evidence is no longer applicable.
Human factors	Have independent reviewers inspect a set of generated audit packages.	Reviewer agreement, time to locate the cause of a blocked result, and unresolved ambiguity.
Security and integrity	Test malformed updates, unauthorized edits, version mismatch, and corrupted evidence records.	Tampering is detected; records remain auditable; failure does not silently pass.

10. Minimum prototype deliverables

- A versioned JSON or database schema for configurations, assumptions, requirements, tests, evidence, and readiness decisions.
- A graph traversal/change-impact implementation with reason-coded outputs.
- A rules-based test selector; defer machine learning until a rules baseline is measured.

- A small benign scenario set: sensor/calibration update, payload swap, software/model revision, stale maintenance or calibration evidence, and conflicting configuration data.
- An exportable evidence package in human-readable form and machine-readable form.
- An evaluation report with full-regression baseline, known false-pass/false-block cases, and reproducibility details.

11. Preliminary art and landscape watchlist

Public records below were surfaced in a first-pass web screen dated 19 October 2026. They are research leads, not a comprehensive search or legal assessment. Use official patent records to verify claims, families, prosecution history, current status, and relevant non-patent literature.

Reference	Search implication
US12475470B1 - Verification and authentication of robots (Google Patents record: https://patents.google.com/patent/US12475470B1/en)	Robot identity, part mismatch, software/configuration/compliance records, digital-twin-style passport, and restricted mode are adjacent. TRACEGATE should focus on change-to-proof-obligation propagation, not identity or ownership transfer.
US20240095354A1 - Assurance model for an autonomous robotic system (https://patents.google.com/patent/US20240095354A1/en)	Multi-layer assurance and probabilistic trust modeling are adjacent. Avoid generic assurance scoring or Bayesian trust as the asserted novelty.
US20230080379A1 - Digital twin for an autonomous vehicle (https://patents.google.com/patent/US20230080379A1/en)	Digital-twin calibration and simulation-to-real data comparison are adjacent. A digital twin may be a test adapter, not the entire differentiator.
US20260127062A1 - Health-aware dynamic contingency management for autonomous UAS (https://patents.google.com/patent/US20260127062A1/en)	Health-aware energy and communications-loss contingency planning is active. Avoid making aircraft contingency control the next standalone concept without a close claim chart.
US10203701B2 - Dynamic task allocation in an autonomous multi-UAV mission (cited in the source AEROSYNC checklist)	Task allocation is already a search-heavy area; reinforces selecting a testing/release layer distinct from fleet planning.

12. Inventor disclosure questions

- What exact problem and technical step did each contributor conceive, and on what date? What was known before that contribution?
- What is the specific dependency representation and the rule for deciding that evidence is no longer applicable after a change?
- How is the targeted test set computed, including safety-critical override rules and uncertainty handling?
- What does READY mean for a defined use envelope, and which conditions make the result stale, restricted, blocked, or revoked?
- What evidence demonstrates an improvement over full regression testing or manual review? If none exists, what benign experiment will produce it?
- Has any version been publicly posted, demonstrated, offered for sale, delivered to a third party, or submitted to a contract sponsor?
- Which package concepts and files are derived from independent contributors, employers, contractors, academic work, or government-funded work?

13. Immediate next actions

9. Resolve the AQUAVANTIS-G UGV/submarine identity discrepancy and freeze the source-of-truth portfolio register.

10. Run a 60-90 minute inventor disclosure session centered on TRACEGATE's change-impact mechanism, not generic autonomy safety.
11. Build a small dependency graph and prototype with 3-5 benign change scenarios before drafting full claims.
12. Commission a claim-focused prior-art search and chart; use a patent practitioner to advise filing strategy and disclosure timing.
13. Only after the mechanism and results are concrete, choose whether to file a provisional or pursue another path with counsel.

Legal and technical caution: these documents are development aids, not legal advice, not a patent application, and not proof of novelty, safety, regulatory compliance, or freedom to operate.